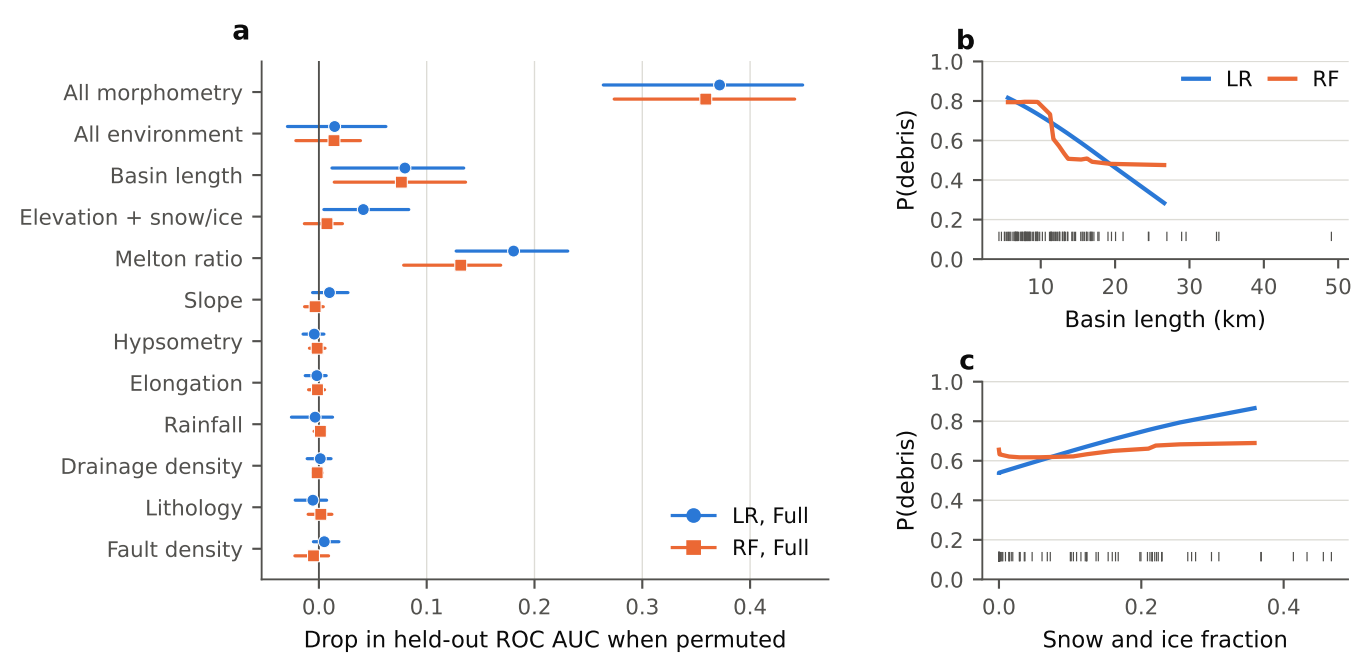
a
All morphometry
All environment
Basin length
Elevation + snow/ice
Melton ratio
Slope
Hypsometry
Elongation
Rainfall
Drainage density
Lithology
Fault density
LR, Full
RF, Full
0.0
0.1
0.2
0.3
0.4
Drop in held-out ROC AUC when permuted
b
LR
RF
P(debris)
1.0
0.8
0.6
0.4
0.2
0.0
10
20
30
40
50
Basin length (km)
c
P(debris)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
Snow and ice fraction